OVERVIEW

aTool-DPO: Multi-Turn DPO for Tool-Augmented LLM

Enhancing TA-LLM Dialogue via Direct Preference Optimization

"Achieving near GPT-4o conversational performance on 8B foundation models."

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Incomplete Queries

Missing Arguments

TA-LLMs trained only with SFT struggle to ask follow-up questions.

Result: Premature or incorrect tool calls.

Failure 1

Out-of-Scope Requests

Unavailable Functionality

Models struggle to reject inappropriate user requests gracefully.

Result: Hallucinated tool invocations.

Failure 2

The SFT Limitation

Absence of Negative Signals

Supervised learning from expert trajectories alone does not explicitly penalize incorrect flows.

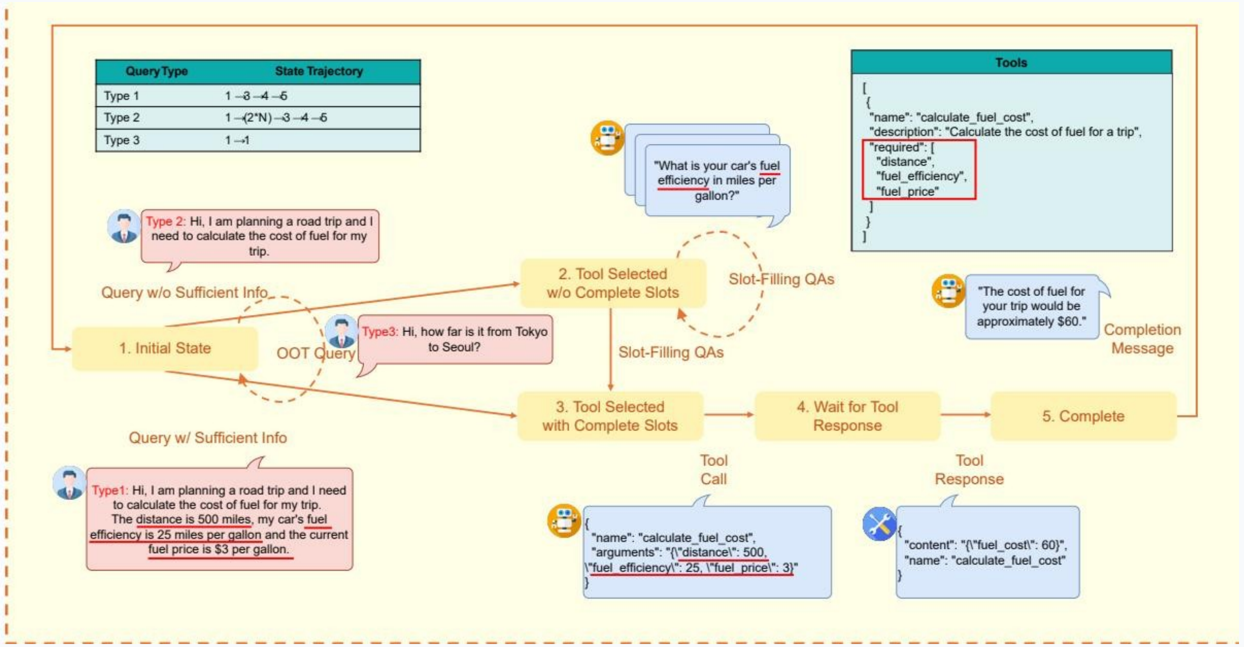
Need: Preference optimization to distinguish right from wrong.

Core Issue

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Internal Unobserved States

1. **Initial State**
   No history. Out-of-scope rejected here.
2. **Tool Selected w/o Complete Slots**
   Slot-filling QA interactions occur here.
3. **Tool Selected w/ Complete Slots**
   Tool and all arguments determined.
4. **Wait for Tool Response**
   Tool call executed, awaiting results.
5. **Complete**
   Results received, message generated.



Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Definition & Example
Type 1	1 → 3 → 4 → 5	**Sufficient Information** Query contains all required arguments. No slot-filling is needed.
Type 2	1 → 2*N → 3 → 4 → 5	**Incomplete Information** Query lacks some arguments. Slot-filling QA occurs N times until complete
Type 3	1 → 1	**Out-of-Scope (Unsupported)** Requested functionality is unavailable. Tool call must be rejected immedia

Automatic Paired Trajectory Construction Without Human Labels

Type 1 Pairs
Sufficient Info

Chosen: 1 → 3 → 4 → 5

Rejected: 1 → 2 → 3 → 4 → 5

Teaches:
Prevent redundant slot-filling.

~5,181 Pairs

Type 2 Pairs
Incomplete Info

Chosen: 1 → 2 → 3 → 4 → 5

Rejected: 1 → 3 → 4 → 5

Teaches:
Prevent slot hallucination.

~5,181 Pairs

Type 3 Pairs
Out-of-Scope

Chosen: 1 → 1

Rejected: 1 → 3 → 4

Teaches:
Tool call rejection.

~1,129 Pairs

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	**0.833**	0.609	0.575	0.614
	SFT-only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
	SFT + DiaTool-DPO	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
	SFT-only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
	SFT + DiaTool-DPO	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
	SFT-only	0.843	**0.957**	0.639	0.826	0.844	0.816
	SFT + DiaTool-DPO	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance

Dramatic Improvements

Slot-filling Accuracy Jump

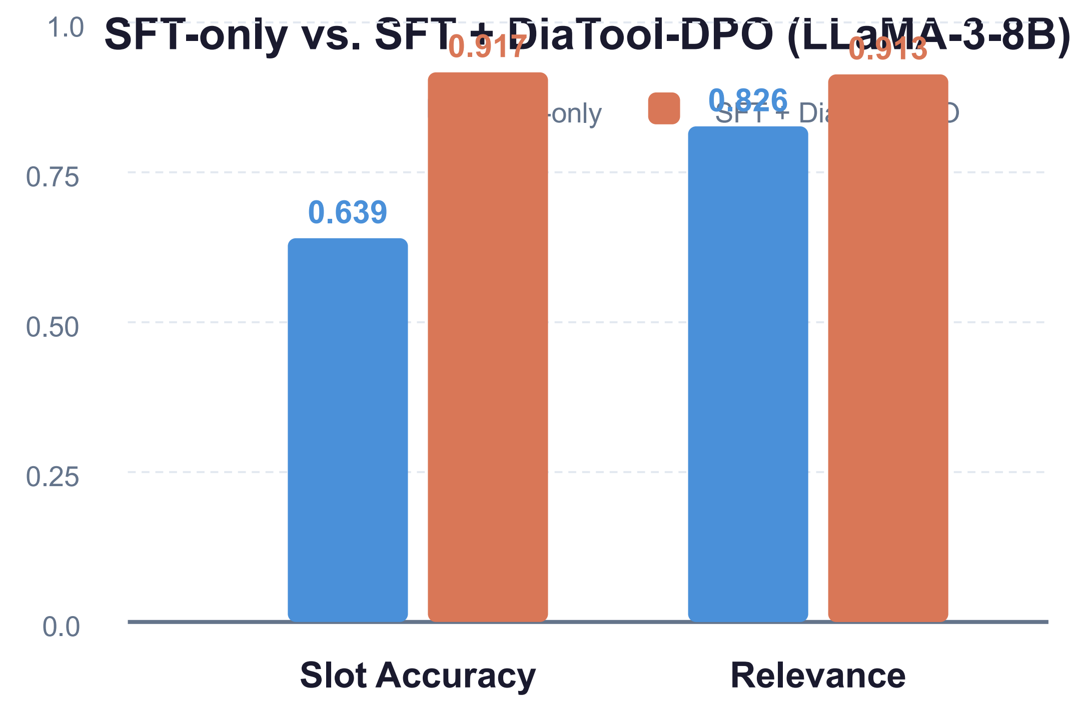
Improved from 0.639 to **0.917** (+43.5%)

Tool Call Rejection Jump

Improved from 0.826 to **0.913** (+10.5%)

The Power of Complementary DPO

Demonstrates DPO successfully complements SFT for critical dialogue skills on open models.



Key Takeaways: RL-Based Dialogue Control

MDP formulation enables systematic pairing

Defining internal states allows automatic trajectory construction without human labeling

DPO strongly complements SFT

Direct Preference Optimization explicitly penalizes hallucinated flows and redundant queries

Language-agnostic & Scalable

Applicable to open-source foundation models, achieving near-GPT-4o conversational capability